

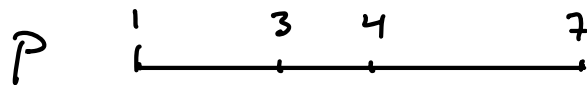
Class 37 Darboux's Theorem

Goals

1. Know what it means for a function to be Riemann integrable on an interval $[a,b]$.
2. Use Darboux's Theorem to calculate the exact value of a definite integral.

Key idea: If f is integrable on the interval $[a,b]$ and P is any partition of $[a,b]$, then

Example: Find lower and upper bounds for the integral $\int_1^7 2-x \, dx$ given by the partition $P = \{1, 3, 4, 7\}$



interval	$M_i(f)$	$m_i(f)$	Δx_i	$M_i \Delta x_i$	$m_i \Delta x_i$
$[1, 3]$					

$[3, 4]$

$[4, 7]$

If Q is a refinement of P , the lower and upper sums for Q form a tighter approximation for the integral.

Now let us consider all possible partitions of the interval $[a,b]$.

We denote the supremum of all the lower sums by $L(f)$ and the infimum of all the upper sums by $U(f)$.

Definition: A bounded function f defined on interval $[a, b]$ is Riemann integrable if $L(f) = U(f)$.

But it is too hard to calculate $U(f)$ and $L(f)$ by considering all partitions. We need a theorem to find the values a different way.

Darboux's Theorem: If f is any bounded function that is Riemann integrable on $[a, b]$ and if $\{P_n : n \in \mathbb{N}\}$ is a sequence of partitions of $[a, b]$ such that every subinterval of P_n has width less than or equal to $\frac{1}{n}$, then $U(f)$ and $L(f)$ both satisfy

Example: Use Darboux's Theorem to evaluate $\int_1^6 x + 1 dx$

Partition the interval $[1,6]$ into subintervals of width $\frac{1}{n}$

upper height =

lower height =

width of each rectangle =

area of upper rectangle =

area of lower rectangle =

$U(f, P_n) =$

$L(f, P_n) =$

$$\begin{aligned} U(f) &= \lim_{n \rightarrow \infty} U(f, P_n) = \lim_{n \rightarrow \infty} 22.5 + \frac{5}{2n} = 22.5 \\ L(f) &= \lim_{n \rightarrow \infty} L(f, P_n) = \lim_{n \rightarrow \infty} 22.5 - \frac{5}{2n} = 22.5 \end{aligned} \quad \left. \vphantom{\lim_{n \rightarrow \infty}} \right\} = \int_1^6 x + 1 dx$$

